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Transactions on Terahertz Science and Technology - Decision onManuscript ID T-TST-REG-01-2026-00004

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Dear Ms. Liu:

Manuscript ID T-TST-REG-01-2026-00004 entitled "Concurrent Terahertz Image Differentiation and Integration Processing via Polarization-Multiplexed Metasurface", which you submitted to the IEEE Transactions on Terahertz Science and Technology, has been reviewed. The reviewer(s)' comments are included at the bottom of this email. *If you do not see the comments or if you cannot read an attached reviewer file, it is very important that you contact our Editorial Assistant, Sharri Shaw, at* thzedit@outlook.com.

The reviewer(s) have recommended **possible publication under the condition of a major revision** of your manuscript. Therefore, I invite you to respond to the reviewer(s)' comments and revise your manuscript accordingly.

To revise your manuscript:

- Please **respond to all comments made by the reviewer(s)**. Your response(s) should be implemented in the revised manuscript and argued in the author's response letter.
- Please note that you will not be able to make revisions to the originally submitted version of the manuscript. Instead, revise your manuscript using a Word/LaTex processing program and save it on your computer. Please **highlight the changes to your manuscript** within the document by using the track changes mode in Word or by using bold or coloured text.
- When submitting your revised manuscript, you should respond to the comments made by the reviewers attaching a separate PDF file (**author's response letter**). You can use this letter to argue your responses and document any changes you make to the original manuscript. In order to expedite the processing of the revised manuscript, please be as specific as possible in your response to the reviewer(s). Please upload this letter as supplementary material and not as the cover letter to the Editors.
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Because we are trying to facilitate timely publication of manuscripts submitted to the IEEE Transactions on Terahertz Science and Technology, **your revised manuscript should be submitted by 18-Mar-2026**. If you are unable to submit by this date, please contact our Editorial Assistant for options.

Once again, thank you for submitting your manuscript to the Transactions on Terahertz Science and Technology and I look forward to receiving your revision.

Sincerely,
Dr. Nuria Llombart
Track Editor, Transactions on Terahertz Science and Technology
n.llombartjuan@tudelft.nl

AE Comments to Author:
Associate Editor: Withayachumnankul, Withawat
Comments to the Author:
The Reviewers are in general satisfied with the manuscript. There are some questions that must be clarified. Please implement ALL changes in the paper.

On top of that I notice the simulation in Fig. 2 is at 632 nm. It's better to perform that at a terahertz frequency given the authors plan to publish in this Transactions.

Reviewer(s)' Comments to Author:
Reviewer: 1

Comments:
In the manuscript “Concurrent Terahertz Image Differentiation and Integration Processing via Polarization-Multiplexed Metasurface”, the authors propose a terahertz image processor based on a polarization-multiplexed metasurface capable of performing differentiation and integration concurrently. Both simulation and experimental results demonstrate excellent terahertz image processing performance. The proposal is novel and the workload is substantial. It is recommended to publish it after minor revisions.
1. The authors report a metasurface based on molybdenum in the manuscript, but do not seem to discuss the sample preparation method in detail—was it fabricated via laser direct writing or semiconductor processes? It is recommended that the authors provide further clarification.
2. In the measurement setup shown in Fig. 3, what is the scanning resolution of the detector? The experimental results in Fig. 5 appear quite smooth.
3. The metasurface presented by the authors essentially functions as a spatially amplitude-modulated filter. What are its advantages compared to phase-modulated devices or devices with amplitude modulation in momentum space? For example, as reported in Appl. Phys. Lett. 126, 081702 (2025); Opto-Electron Sci 3, 230052 (2024); and Opto-Electronic Advances 7, 240075 (2024).
4. The authors control the transmission amplitude response by spatially orienting the meta-atoms. Have they investigated the influence of the incident angle—that is, the amplitude response in momentum space?

Additional Questions:
Would you nominate this paper for the Terahertz Transactions Best Paper Prize? If “Yes” Briefly discuss your reasons.: No

The Transactions on Terahertz Science and Technology accepts new and original articles describing significant work and/or ideas not found elsewhere in the literature. Is this work a new and original contribution in the areas of Terahertz science, technology or applications?: Yes

The Transactions does not accept incremental submissions. After considering all prior work published by the authors and others, would you characterize this submission as incremental?: No

All related previous work by the authors must be referenced, even if it was presented in conference. Is previous work by the authors properly referenced?: Yes

If you are suggesting additional references they must be entered in the text box provided. All suggestions must include full bibliographic information plus a DOI.
: No

The Transactions generally requires that theory, measurements, and simulations be supported by careful comparisons. Do the theory, measurements, and simulations presented meet this requirement?: Yes

Please provide a ranking (0-10) of the significance of this work.: 9

Reviewer: 2

Comments:
This manuscript reports a polarization-multiplexed metasurface integrated into a 4f terahertz imaging system to perform concurrent image differentiation and integration. Experimental validation using a THz-TDS system shows reasonable agreement with simulations for both square and hexagram test objects. I think this is an interesting experimental demonstrations of concurrent differentiation and integration for terahertz images using a single metasurface layer. Overall, the topic is timely and relevant to the readership of the journal, particularly those working on THz imaging systems and metasurfaces. Following are few comments, which need to be addressed before accepting this article.

1. While differentiation and integration are performed simultaneously in two polarization channels, the manuscript would benefit from a clearer explanation of how the outputs are practically separated in the experiment. For example, specify whether polarization analyzers were used at detection and how polarization cross-talk affects performance.
2. The metasurface transmission is broadband, but the image processing results are effectively shown at selected frequencies. Please clarify which frequency or frequency range is used to reconstruct the images and how sensitive the results are to frequency variation.
3. Although the authors state that this is the first demonstration of concurrent differentiation and integration in the THz band, a more explicit comparison table or paragraph contrasting this work with existing THz image processing approaches would improve context and impact.
4. Some figure captions are long but still lack clarity on experimental conditions, especially Fig. 5 and Fig. 7. Please specify the polarization state used for each result directly in the captions.
5. There are several grammatical and typographical issues that should be corrected. For example, “tntegration” in the conclusion, repeated words, and inconsistent spacing in equations and symbols.

Additional Questions:
Would you nominate this paper for the Terahertz Transactions Best Paper Prize? If “Yes” Briefly discuss your reasons.: No

The Transactions on Terahertz Science and Technology accepts new and original articles describing significant work and/or ideas not found elsewhere in the literature. Is this work a new and original contribution in the areas of Terahertz science, technology or applications?: I think this idea is new in terahertz region but not in other wavelegth region.

The Transactions does not accept incremental submissions. After considering all prior work published by the authors and others, would you characterize this submission as incremental?: No

All related previous work by the authors must be referenced, even if it was presented in conference. Is previous work by the authors properly referenced?: Yes

If you are suggesting additional references they must be entered in the text box provided. All suggestions must include full bibliographic information plus a DOI.
: No

The Transactions generally requires that theory, measurements, and simulations be supported by careful comparisons. Do the theory, measurements, and simulations presented meet this requirement?: I think so.

Please provide a ranking (0-10) of the significance of this work.: 6

Reviewer: 3

Comments:
This paper proposes a metasurface applied in a 4f system to achieve edge enhancement and denoising, essentially functioning as a polarization-sensitive filter. The metasurface consists of horizontally oriented patches at the center, which allow vertically polarized waves to pass through, and vertically oriented patches arranged in a circular ring around the center, which transmit horizontally polarized waves. This is what the authors refer to as “functional concurrency.” In a 4f system, the field at the focal plane is equivalent to the frequency domain of the Fourier transform, and placing the metasurface at this focal plane enables polarization filtering.

I encourage the authors to clarify the novelty particularly in following aspects:
(1) Regarding the concept of “Concurrent Processing,” the metasurface is implemented using a single-layer rectangular patch configuration, which appears structurally straightforward and consistent with commonly adopted designs in the literature. The manuscript would benefit from a clearer explanation of how this implementation differentiates itself from existing approaches.

(2) The work can be viewed as extending established optical 4f system concepts into the THz frequency regime. It would strengthen the paper if the authors could more explicitly articulate the unique technical challenges or methodological adaptations required for this frequency translation.

(3) While the low-cost fabrication is certainly a practical advantage, it may be helpful to clarify how this aspect relates to the core scientific contribution of the study, as fabrication affordability alone may not fully constitute methodological innovation.

Additional Questions:
Would you nominate this paper for the Terahertz Transactions Best Paper Prize? If “Yes” Briefly discuss your reasons.: No

The Transactions on Terahertz Science and Technology accepts new and original articles describing significant work and/or ideas not found elsewhere in the literature. Is this work a new and original contribution in the areas of Terahertz science, technology or applications?: Yes

The Transactions does not accept incremental submissions. After considering all prior work published by the authors and others, would you characterize this submission as incremental?: No

All related previous work by the authors must be referenced, even if it was presented in conference. Is previous work by the authors properly referenced?: Yes

If you are suggesting additional references they must be entered in the text box provided. All suggestions must include full bibliographic information plus a DOI.
: No

The Transactions generally requires that theory, measurements, and simulations be supported by careful comparisons. Do the theory, measurements, and simulations presented meet this requirement?: Yes

Please provide a ranking (0-10) of the significance of this work.: 7